



Yeshwantrao Chavan College of Engineering

(An Autonomous Institution affiliated to Rashtrasant Tukadoji Maharaj Nagpur University)

Hingna Road, Wanadongri,

Nagpur - 441 110 NAAC A++

Ph.: 07104-237919, 234623, 329249, 329250 Fax: 07104-232376, Website: www.ycce.edu



Department of Computer Technology

Vision of the Department

To be a well-known centre for pursuing computer education through innovative pedagogy, value-based education and industry collaboration.

Mission of the Department

To establish learning ambience for ushering in computer engineering professionals in core and multidisciplinary area by developing Problem-solving skills through emerging technologies.

Session 2025-2026

Vision: To Become Excellent in Computer vision	Mission: To Make real life project related to computer vision
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Program Educational Objectives of the program (PEO): (broad statements that describe the professional and career accomplishments)

PEO1	Preparation	P: Preparation	Pep-CL abbreviation pronounce as Pep-si-IL easy to recall
PEO2	Core Competence	E: Environment (Learning Environment)	
PEO3	Breadth	P: Professionalism	
PEO4	Professionalism	C: Core Competence	
PEO5	Learning Environment	L: Breadth (Learning in diverse areas)	

Program Outcomes (PO): (statements that describe what a student should be able to do and know by the end of a program)

Keywords of POs:

Engineering knowledge, Problem analysis, Design/development of solutions, Conduct Investigations of Complex Problems, Engineering Tool Usage, The Engineer and The World, Ethics, Individual and Collaborative Team work, Communication, Project Management and Finance, Life-Long Learning

PSO Keywords: Cutting edge technologies, Research

“I am an engineer, and I know how to apply engineering knowledge to investigate, analyse and design solutions to complex problems using tools for entire world following all ethics in a collaborative way with proper management skills throughout my life.” to contribute to the development of cutting-edge technologies and Research.

Integrity: I will adhere to the Laboratory Code of Conduct and ethics in its entirety.



28/8/25

Practical 4

Page No. _____

Date : / /

Aim:- ~~to~~ to apply scale NAF to program
to apply scale invariant feature
transformation on input image.

Theory:-

- Powerful algorithm used in CV
to detect & describe local features

Components of SIFT Algorithm

- Detection of scale space extrema
- Localization of Keypoints
- orientation Assignments
- Keypoint Descriptor creation

Katewar
28/8/25

Ganesh Pandile

Name and Signature of Student and Date
(Signature and Date in Handwritten)



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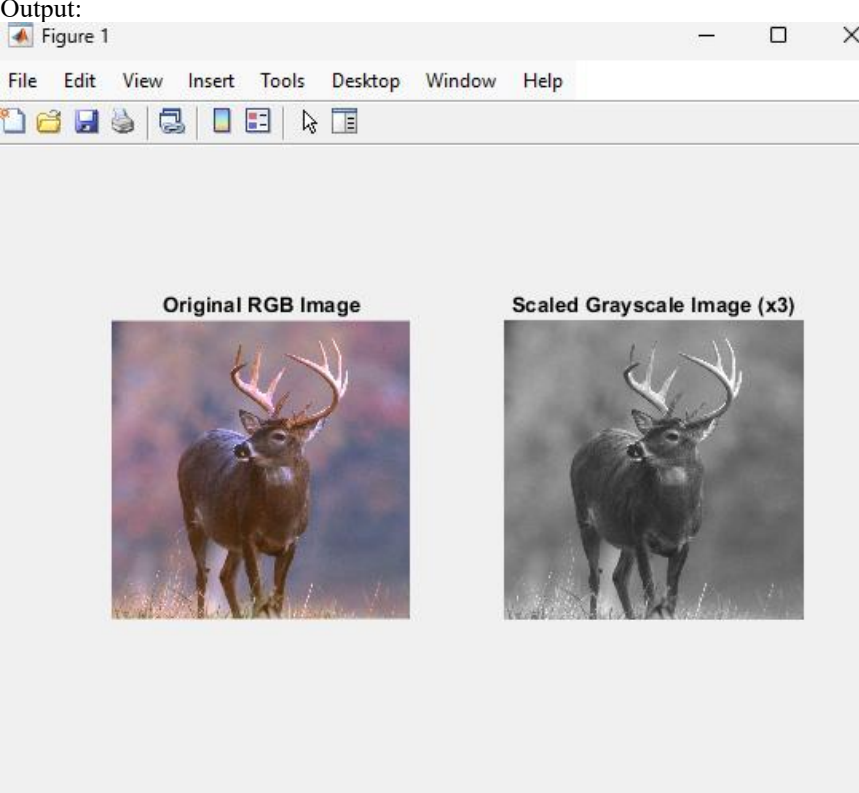
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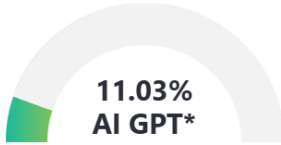
Session	2024-25 (ODD)	Course Name	Computer vision Lab
Semester	5	Course Code	CT
Roll No	46	Name of Student	Ganesh Pandile

Practical Number	4
Course Outcome	Extract meaningful features from images using descriptors such as HOG and SIFT.
Aim	Write a Program to apply Scale Invariant Feature Transform on input image.
Problem Definition	
Theory (100 words)	SIFT (Scale Invariant Feature Transform) is an algorithm in computer vision that detects and describes local features in an image. It finds keypoints that are invariant to scale, rotation, and lighting, making it ideal for object recognition and image matching. Steps : 1. Detect keypoints at different scales using DoG (Difference of Gaussians). 2. Filter and refine keypoints for stability. 3. Assign orientation to make it rotation-invariant. 4. Create descriptors from local gradients around keypoints. 5. Match features between images using these descriptors.
Procedure and Execution (100 Words)	Algorithm: 1. Read the Input Image: - o Load the RGB image from a specified file path. 2. Convert to Grayscale: - o Transform the RGB image into a grayscale image. 3. Set Scaling Factor: - o Define how much to scale the image (e.g., 3 times larger). 4. Calculate New Size: - o Multiply the original grayscale image dimensions by the scaling factor. 5. Resize the Image: - o Use interpolation to scale the grayscale image to the new size. 6. Display Images: - o Show the original RGB image and the scaled grayscale image side by side. 7. Save Scaled Image:



	☐ Write the scaled grayscale image to a new file on disk. 8. Print Confirmation: ☐ Output a message confirming that the image has been saved. Code: <pre> inputImage = imread("C:\Users\student\Desktop\CT_46_CV\deer.jpg"); grayImage = rgb2gray(inputImage); scaleFactor = 3; newSize = round(size(grayImage) * scaleFactor); scaledImage = imresize(grayImage, newSize); figure; subplot(1,2,1); imshow(inputImage); title('Original RGB Image'); subplot(1,2,2); imshow(scaledImage); title(['Scaled Grayscale Image (x' num2str(scaleFactor) ')']); imwrite(scaledImage, 'output_scaled_gray.jpg'); fprintf('Scaled grayscale image saved as output_scaled_gray.jpg\n'); </pre> Output: 
Output Analysis	Displays two images: • Left: Original RGB image • Right: Scaled grayscale image (3× larger) Saves output: • Grayscale scaled image saved as output_scaled_gray.jpg Console message: • Confirms image is saved successfully



	Effect: The image is enlarged and converted to grayscale.
Link of student Github profile where lab assignment has been uploaded	https://github.com/ganeshpandile/Computer-Vision.git
Conclusion	Successfully applied Scale Invariant Feature Transform on input image
Plag Report (Similarity index < 12%)	Your File Content is Human written  11.03% AI GPT*
Date	31/08/2025



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